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Subject: Programming with Java

ASSIGNMENT No : 5

TITLE : Collection Classes and StringBuffer

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Programme

arraylist

```
import java.util.ArrayList;
public class Arraylist
{
    public static void main(String[] args)
    {
        ArrayList<String> fruits = new ArrayList<>();
        fruits.add("Apple");
        fruits.add("banana");
        fruits.add("mongo");
        fruits.add("cherr");
        fruits.add("apple");
        System.out.println("fruits"+fruits);
        fruits.remove("Apple");
        System.out.println(" after removing fruit"+fruits);
        System.out.println("size: "+fruits.size());
        System.out.println("contain "+fruits.size());
    }
}
```

Stringbuffer

```

public class StringBufferexample
{
    public static void main(String[] args)
    {
        StringBuffer sb=new StringBuffer("hello ");
        sb.append("world");
        System.out.println(";after append "+ sb);
        sb.insert(6, "java");
        System.out.println("after insert "+ sb);
        sb.replace(0,5,"hi");
        System.out.println("after replacing "+ sb);
        sb.delete(2, 5);
        System.out.println("after delate "+ sb);
        sb.reverse();
        System.out.println("after after reverse "+ sb);
    }
}

```

Vector

```

import java.util.Vector;
public class vector
{
    public static void main(String[] args)
    {
        Vector<Integer> number = new Vector<>();
        number.add(100);
        number.add(200);
        number.add(300);
        number.add(400);
        number.add(500);
        System.out.println("numbers are "+number);
        number.remove(4);
        System.out.println(" after removing "+number);
        System.out.println(" size"+number.size());
        System.out.println("contain numbers "+number.get(0));
    }
}

```

Output

Arraylist

```

PS C:\CS-JAVA> javac Arraylist.java
PS C:\CS-JAVA> java Arraylist
fruits[Apple, banana, mongo, cherr, apple]
after removing fruit[banana, mongo, cherr, apple]
size: 4
contain 4
PS C:\CS-JAVA> 

```

Strinbuffer

```
PS C:\CS-JAVA> javac StringBufferexample.java
PS C:\CS-JAVA> java StringBufferexample
;after append hello world
after insert hello javaworld
after replacing hi javaworld
after delate hivaworld
after after reverse dlrowavih
PS C:\CS-JAVA>
```

Veector

```
PS C:\CS-JAVA> javac vector.java
PS C:\CS-JAVA> java vector
numbers are [100, 200, 300, 400, 500]
after removing [100, 200, 300, 400]
size4
contain numbers 100
PS C:\CS-JAVA>
```

Conclusion

The Java application successfully demonstrates the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation. ArrayList and Vector allow efficient storage and modification of collections of data, while StringBuffer provides methods for performing mutable string operations. This experiment helps in understanding the differences between these classes and their applications in Java programming for managing dynamic data efficiently.